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MAR-APR-2024 SUMMER EXAMINATION

B.Tech. CBCS

Sub. Name: Switchgear and Protection

Sub. Code: 67517/83779/83974

Day and Date: MAY ,04-05-2024 **Total Marks: 70**

Time: 10:30 AM To 01:00 PM

Instructions:
1. **All questions are compulsory**
2. **Assume suitable data wherever necessary and mention it boldly**
3. **Draw neat labbelet diagrams wherever necessary**
4. **Figures to the right indicate full marks**

Q1) Attempt the following questions **[12]**

a. Explain High resistance and Current zero interruption in detail. **[6]**

b. Explain Air break circuit breaker in detail. **[6]**

OR

Explain voltage current characteristics of arc.

Q2) Attempt the following questions **[12]**

a. Write note on induction disc type relay in detail. **[6]**

b. Explain attracted armature type relay in detail. **[6]**

OR

Explain Microprocessor based over current relay in detail.

Q3) Attempt the following questions **[12]**

a. Explain directional relay in detail. **[6]**

b. Explain Feeder relay protection scheme in detail. **[6]**

OR

Write note on Plug setting & Time setting multiplier.

Q4) Attempt the following questions **[12]**

a. Explain circulating current & opposed voltage principle of differential relay. **[6]**

b. Explain transmission line relay protection scheme. **[6]**

OR

Explain working of percentage differential relay.

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Q5) Attempt the following questions **[12]**

a. Explain protection against magnetizing inrush current in transformer. **[6]**

b. Draw a neat sketch of Buchholz relay & explain its working principle. State its application. **[6]**

OR

Explain problems associated with percentage differential protection.

Q6) Attempt the following questions **[10]**

a. Explain in detail microprocessor based mho relay. **[5]**

b. Write a note on selection of distance relays. **[5]**

OR

Explain generator protection for earth fault protection & excitation failure.

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